(TR-104) PYTHON WITH AI/ML

Training Day 51 Report

Date: 23 March 2026

Topic Covered: Streamlit UI Development, ML Integration & Deployment

The fifty-first day of training focused on developing interactive Streamlit applications, exploring advanced widgets and UI components, integrating machine learning models into Streamlit, and deploying applications on Streamlit Community Cloud.

- Built a User Registration Form using st.form with proper input validation handling.
- Designed structured user interfaces using Streamlit columns for improved layout organization.
- Explored multiple Streamlit widgets including checkbox, color picker, feedback/rating component, multiselect, pills, radio buttons, toggle, date input, and time input.
- Captured user queries and messages using Streamlit chat input functionality.
- Applied custom styling and formatting using st.markdown.
- Implemented button-based interactions inside Streamlit applications.
- Added image download functionality using st.download_button.
- Generated CSV reports dynamically and enabled file download functionality.
- Integrated external navigation using link buttons.
- Used st.cache_data for efficient dataset loading and performance optimization.
- Used st.cache_resource for optimized loading of machine learning models and reusable resources.
- Measured execution time to analyze application performance improvements.
- Implemented st.session_state to preserve data and counters across reruns.
- Integrated sentiment analysis using Hugging Face Transformers pipeline inside Streamlit.

- Loaded and used pretrained transformer models for inference.
- Optimized ML model usage using caching techniques.
- Built an interactive Data Explorer application using Pandas for data processing and analysis.
- Implemented filtering functionality using sliders and multiselect widgets.
- Processed and reshaped datasets dynamically for interactive analysis.
- Enabled editable data tables using st.data_editor.
- Visualized data interactively using Altair line charts.
- Successfully deployed the YouTube Keyword Extractor application on Streamlit Community Cloud.
- Tested the deployed application to verify proper functionality and deployment workflow.
- Strengthened understanding of frontend development and deployment workflows for AI-powered applications.
- **GitHub Link:** https://github.com/JaspinderKaurWalia26/youtube_keyword
- **Blog Link:** https://medium.com/@jaspinderkaurwalia855/building-interactive-forms-in-streamlit-a-beginner-friendly-guide-f607d7a81b25